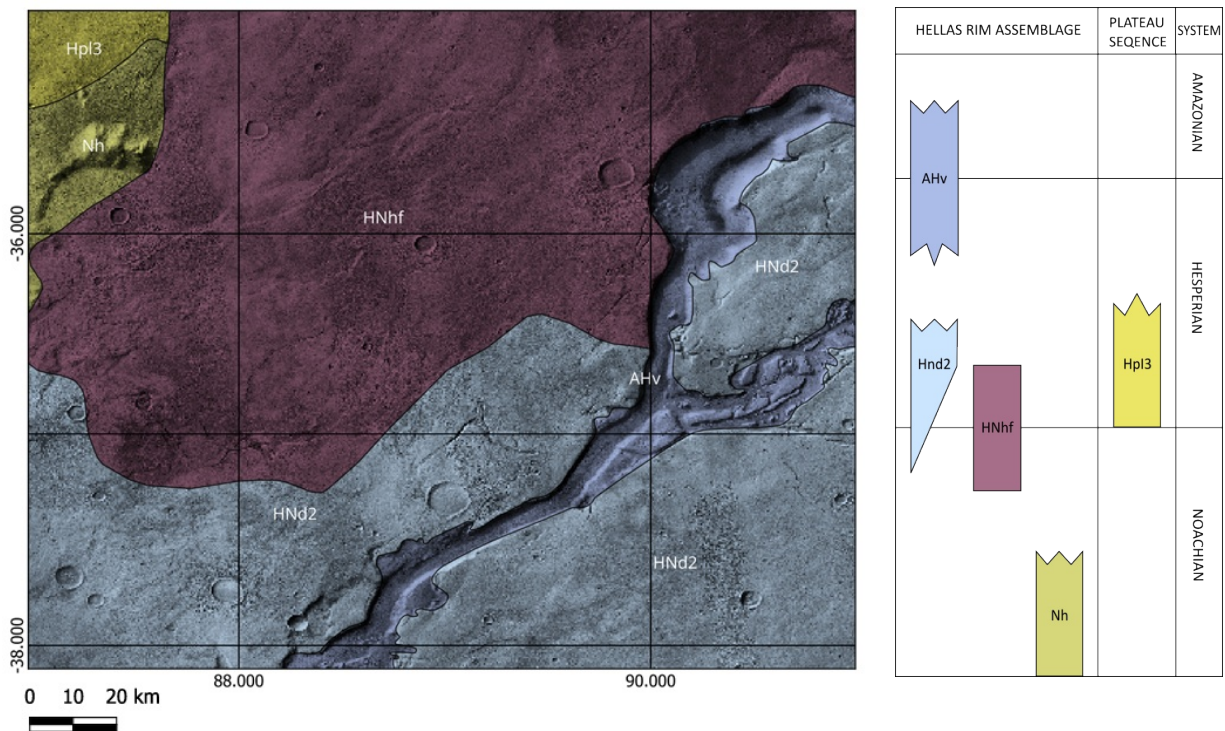


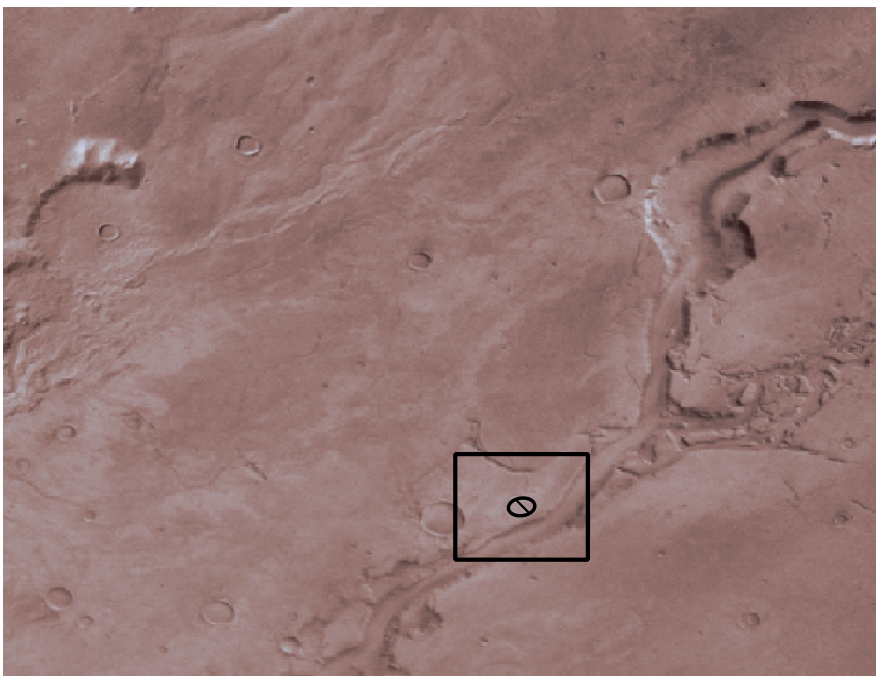
Geologic Mapping and Trajectory Planning in the Dao Vallis Region, Mars

Geologic Investigations

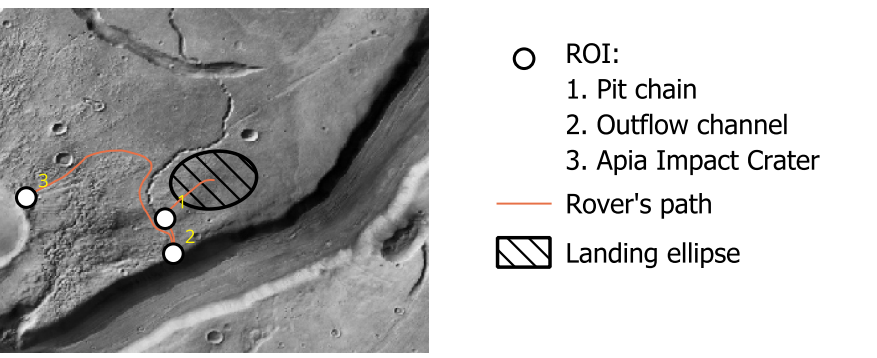


- AHv** - forms the smooth floors of large valleys (like Dao, Niger, and Harmakhis) and deep depressions. Interpretation: Deposits from catastrophic groundwater outbreaks and debris flows, with subsequent fluvial activity.
- HNd2** - forms broad, uneven, moderately cratered plains that are cut by numerous channels. Interpretation: Alluvium and probably volcanic/eolian materials that were highly dissected by fluvial (water) erosion.
- HNhf** - moderately cratered, layered, and highly dissected materials surrounding the Hadriaca Patera volcano, extending for over 400 km. Interpretation: Pyroclastic flows erupted from Hadriaca Patera that were later dissected by fluvial processes and groundwater sapping.
- Hpl3** - forms smooth, flat to undulating, relatively featureless plains with moderate cratering. Interpretation: Thick fluvial, eolian (wind), and volcanic deposits that buried most of the older underlying rocks.
- Nh** - forms rugged, high-relief, densely cratered terrain featuring numerous isolated mountain massifs and channels. Interpretation: Uplifted crustal material and ejecta from the massive Hellas impact, which later underwent structural and erosional modifications.

General site plan with the designated drop-off zone



Context location map of ROI



Coordinate System: Mars 2000 Equirectangular
Datum: Mars IAU 2000
Imagery: MRO Context Camera (CTX)